trail-creek-exploratory

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Packages

```
library(tidyverse)
library(here)
library(janitor)
library(snakecase)
library(scales)
library(readxl)
library(patchwork)
library(ggridges)
library(lme4)
library(lmerTest)
library(knitr)
library(rmarkdown)
library(viridis)
library(viridisLite)
library(wesanderson)
library(ragg)
```

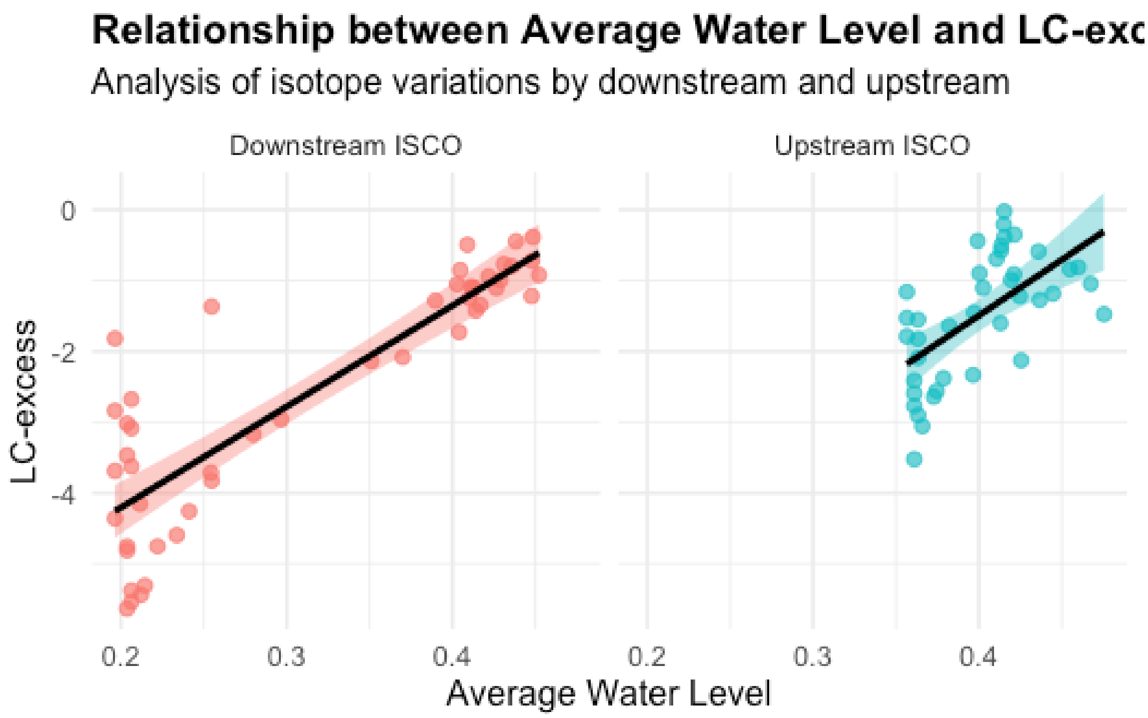
Data

```
compiled_isotope  <-  read_csv(here("data",  "compiled.csv"),  show_col_types  =
FALSE) |>
  clean_names() |> # lowercase and underscores in column names
  rename(sample = sample_name, # renaming columns to be shorter
         delta_d = processed_delta_d,
         delta_d_std = processed_delta_d_stdev,
         delta_o = processed_delta_18o,
         delta_o_std = processed_delta_18o_stdev) |>
  mutate(date = mdy(date)) |>  # telling r the date format
  filter(location %in% c("utc", "tc"))

site_order <- c("tc_isco", "tc_1", "tc_15", "tc_2", "tc_3",
                "utc_isco", "tc_ds", "tc_mid", "tc_us")

trail_creek <- compiled_isotope |>
```

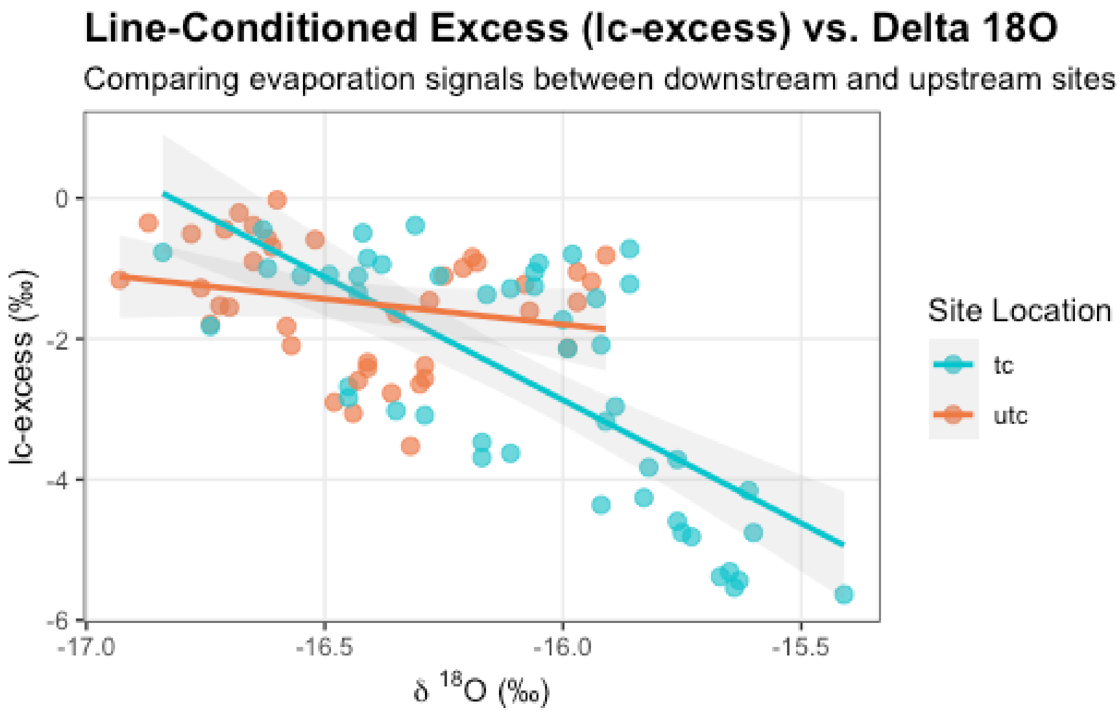
```r
  filter(location %in% c("utc", "tc")) |>
  mutate(site = factor(site, levels = site_order))

tc_isotope <- trail_creek |>
  filter(location %in% c("tc"))

utc_isotope <- trail_creek |>
  filter(location %in% c("utc"))

gradient_tc <- read_csv(here("data", "tc_gradient_data.csv"))

metadata <- read_csv(here("data", "metadata.csv"))

# Downstream Trail Creek
tc_data     <-     read_csv(here("data",     "TrailCreek_2026-07-23-15.csv"),
show_col_types = FALSE) |> # reading in data
  clean_names() |> # lowercase and underscores in column names
  rename(date_time = date_time_mdt, # renaming columns to be shorter
         dp = differential_pressure_k_pa,
         ap = absolute_pressure_k_pa,
         temp = temperature_c,
         water_level = water_level_m,
         bp = barometric_pressure_k_pa) |>
  mutate(date_time_parsed = mdy_hms(date_time), # telling r the date format
         date = as_date(date_time_parsed)) |> # making a date column
  group_by(date) |>
   summarise(avg_dp  =  mean(dp,  na.rm  =  TRUE),  #  averaging  the  data  based  on
the day
            avg_ap = mean(ap, na.rm = TRUE),
            avg_temp = mean(temp, na.rm = TRUE),
            avg_water_level = mean(water_level, na.rm = TRUE),
            avg_bp = mean(bp, na.rm = TRUE),
            total_readings = n(), # the amount of data it used to average
            .groups = "drop") |>
  mutate(site = "tc") # creating a new column of the site

# Upstream Trail Creek
utc_data  <-  read_csv(here("data",  "UTC_2026-07-23-15.csv"),  show_col_types  =
FALSE) |> # reading in data
  clean_names() |> # lowercase and underscores in column names
  rename(date_time = date_time_mdt,  # renaming columns to be shorter
         dp = differential_pressure_k_pa,
         ap = absolute_pressure_k_pa,
```

```
         temp = temperature_c,
         water_level = water_level_m,
         bp = barometric_pressure_k_pa) |>
  mutate(date_time_parsed = mdy_hms(date_time), # telling r the date format
         date = as_date(date_time_parsed)) |> # making a date column
  group_by(date) |>
   summarise(avg_dp  = mean(dp,  na.rm  = TRUE),   # averaging the data based on
the day
            avg_ap = mean(ap, na.rm = TRUE),
            avg_temp = mean(temp, na.rm = TRUE),
            avg_water_level = mean(water_level, na.rm = TRUE),
            avg_bp = mean(bp, na.rm = TRUE),
            total_readings = n(), # the amount of data it used to average
            .groups = "drop") |>
  mutate(site = "utc") # creating a new column of the site

tc_utc_data <- merge(tc_data, utc_data, all = TRUE) # merging the tc and utc data

isco_data <- merge(
  x = tc_utc_data,
  y = compiled_isotope,
  by.x = c("site", "date"), # columns in tc_utc_data
  by.y = c("location", "date") # matching columns in compiled_isotope
)
```

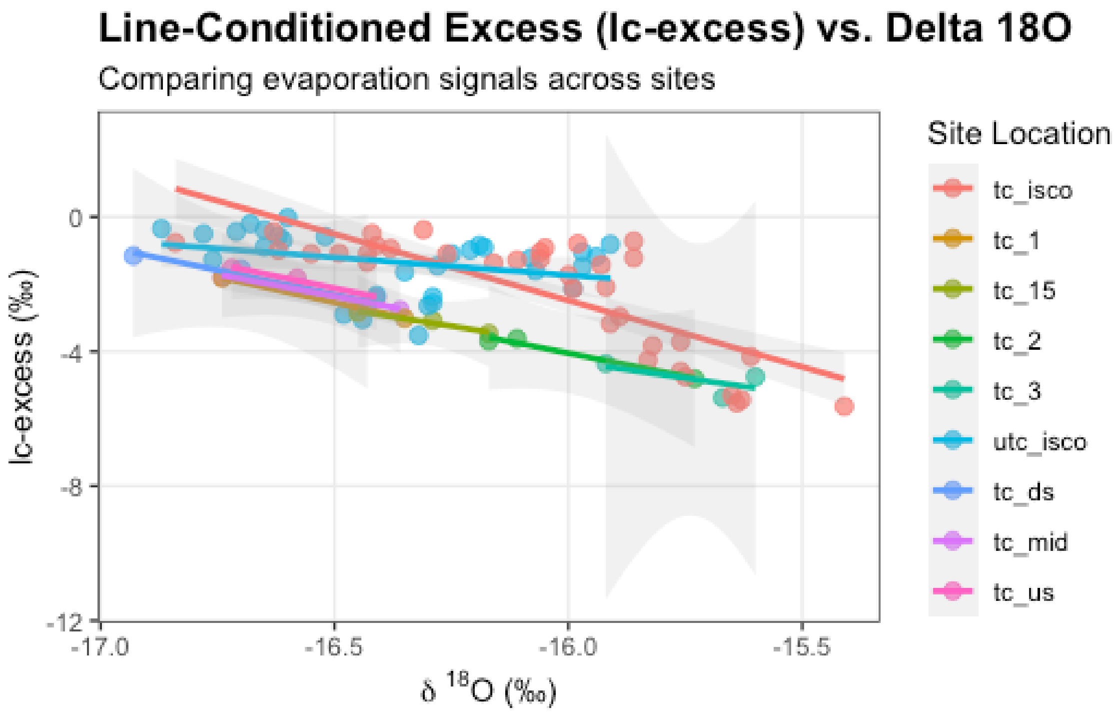
```
ggplot(data = isco_data, aes(x = avg_water_level, y = lc_excess, color = site)) +
  geom_point(size = 2, alpha = 0.7) +
  geom_smooth(method = "lm", aes(fill = site), color = "black", se = TRUE) +
  labs(
    title = "Relationship between Average Water Level and LC-excess",
    subtitle = "Analysis of isotope variations by downstream and upstream",
    x = "Average Water Level",
    y = "LC-excess"
  ) +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(face = "bold", size = 14),
        legend.position = "none") +
  facet_wrap(~site, labeller = as_labeller(c("tc" = "Downstream ISCO",
                                             "utc" = "Upstream ISCO")))
```




```
ggplot(data = trail_creek, aes(x = delta_o, y = lc_excess, color = location)) +
  geom_point(size = 2.5, alpha = 0.7) +
  geom_smooth(method = "lm", se = TRUE, alpha = 0.15, size = 1) +
  scale_color_manual(values = c("tc" = "turquoise3", "utc" = "sienna2")) +
  labs(
    title = "Line-Conditioned Excess (lc-excess) vs. Delta 18O",
     subtitle = "Comparing evaporation signals between downstream and upstream
sites",
    x = expression(paste(delta, " "^18, "O (\u2030)")), # Formats as δ18O (‰)
    y = "lc-excess (\u2030)",
    color = "Site Location") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        legend.position = "right",
        panel.grid.minor = element_blank())
```

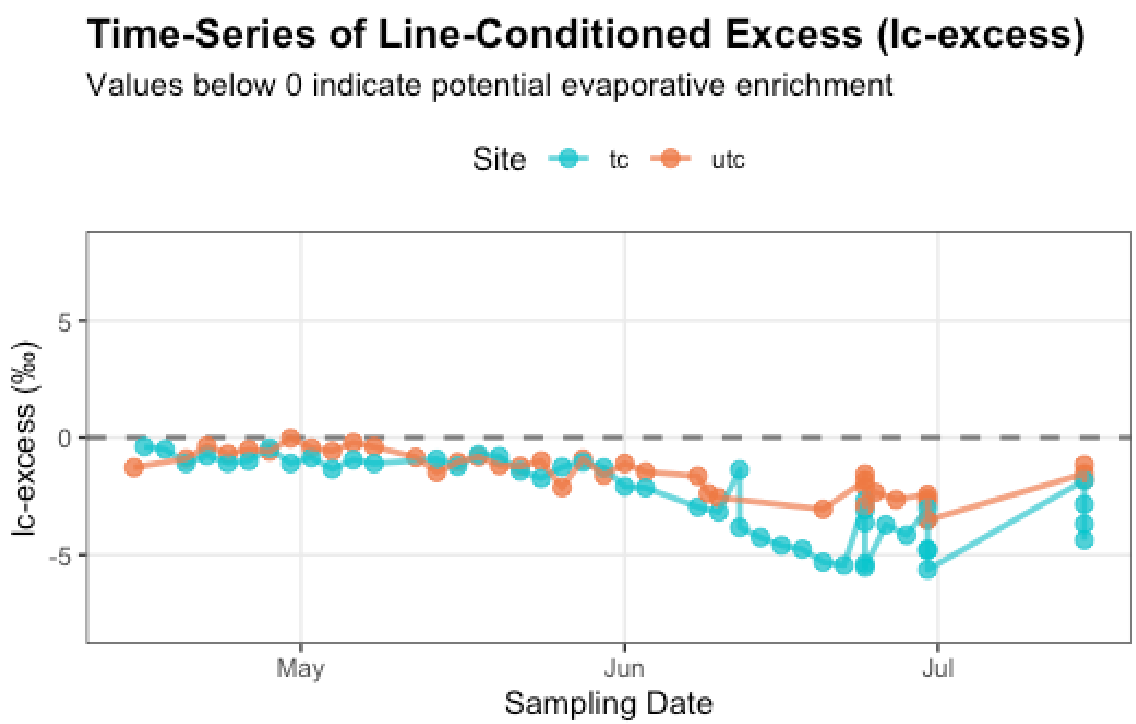



```r
ggplot(data = trail_creek, aes(x = delta_o, y = lc_excess, color = site)) +
  geom_point(size = 2.5, alpha = 0.7) +
  geom_smooth(method = "lm", se = TRUE, alpha = 0.15, size = 1) +
  labs(
    title = "Line-Conditioned Excess (lc-excess) vs. Delta 18O",
    subtitle = "Comparing evaporation signals across sites",
    x = expression(paste(delta, " "^18, "O (\u2030)")), # Formats as δ18O (‰)
    y = "lc-excess (\u2030)",
    color = "Site Location") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        legend.position = "right",
        panel.grid.minor = element_blank())
```




```
ggplot(data = trail_creek,
       aes(x = date,
           y = lc_excess,
           color = location,
           group = location)) +
  geom_hline(yintercept = 0,
             linetype = "dashed",
             color = "gray50",
             size = 0.8) +
  geom_point(size = 2.5,
             alpha = 0.8) +
  geom_line(size = 1,
            alpha = 0.7) +
  scale_color_manual(values = c("tc" = "turquoise3",
                                "utc" = "sienna2")) +
  labs(title = "Time-Series of Line-Conditioned Excess (lc-excess)",
       subtitle = "Values below 0 indicate potential evaporative enrichment",
       x = "Sampling Date",
       y = "lc-excess (\u2030)",
       color = "Site") +
  ylim(-8,8) +
  theme_bw() +
```

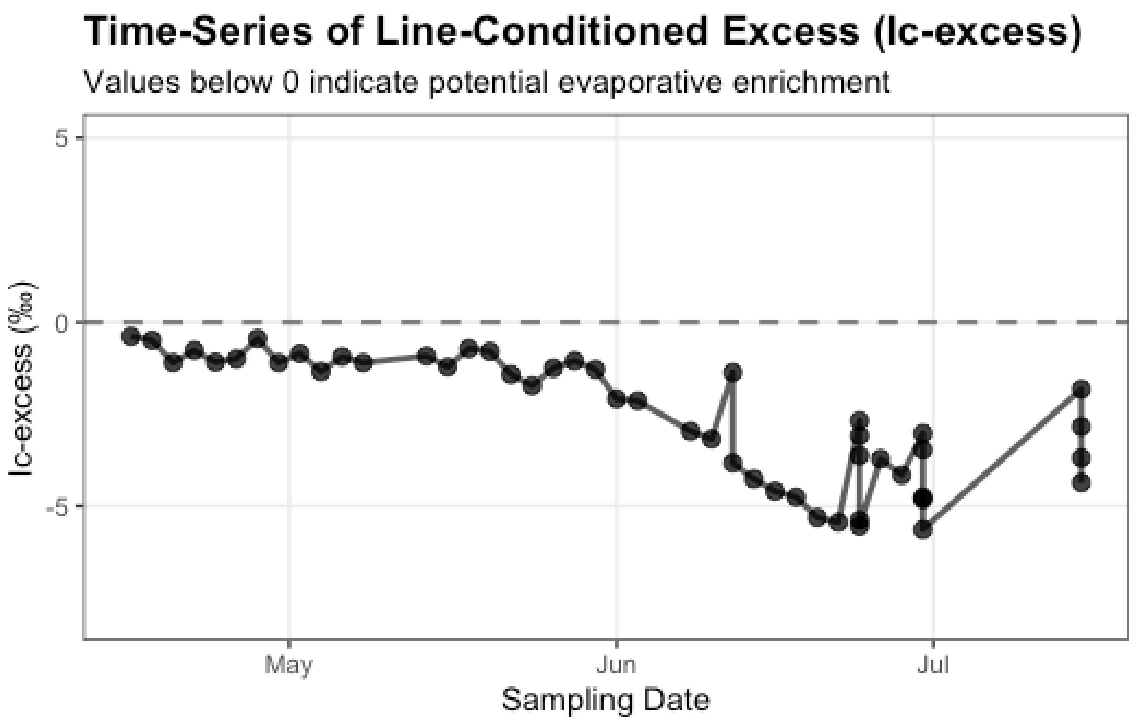
```r
  theme(plot.title = element_text(face = "bold", size = 14),
        panel.grid.minor = element_blank(),
        legend.position = "top")
```



Time-Series of Line-Conditioned Excess (lc-excess)

```r
ggplot(data = tc_isotope,
       aes(x = date,
           y = lc_excess)) +
  geom_hline(yintercept = 0,
             linetype = "dashed",
             color = "gray50",
             size = 0.8) +
  geom_point(size = 2.5,
             alpha = 0.8) +
  geom_line(size = 1,
            alpha = 0.7) +
  labs(title = "Time-Series of Line-Conditioned Excess (lc-excess)",
       subtitle = "Values below 0 indicate potential evaporative enrichment",
       x = "Sampling Date",
       y = "lc-excess (\u2030)",
       color = "Site") +
```

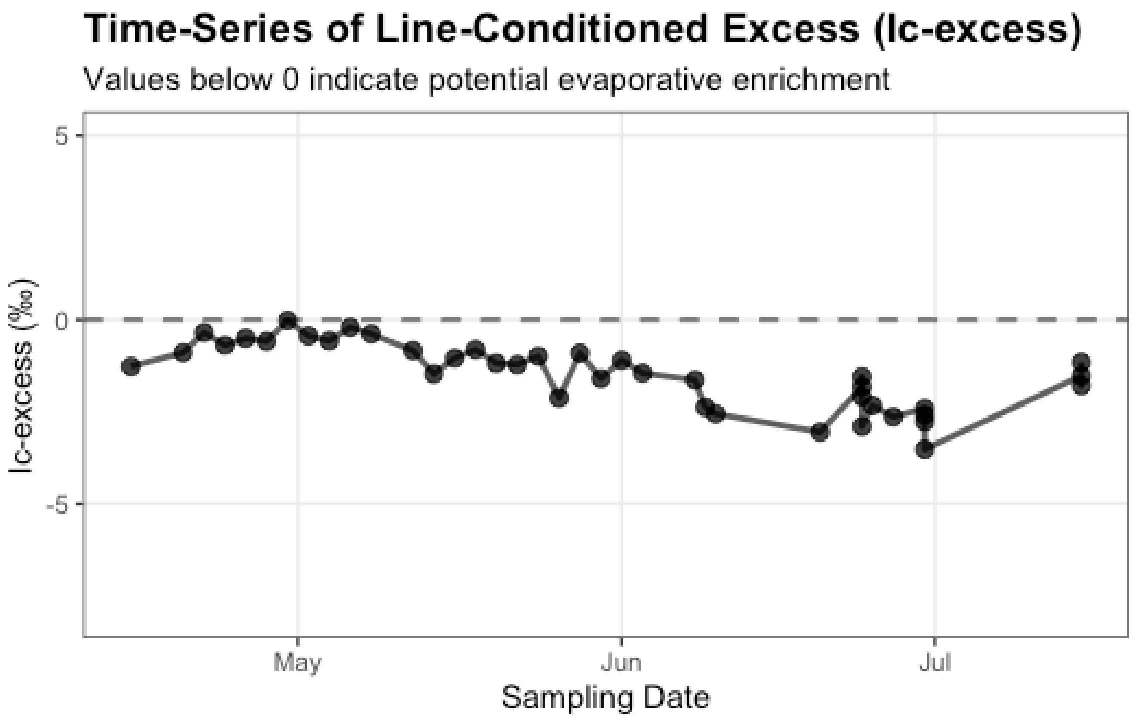
```
  ylim(-8,5) +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        panel.grid.minor = element_blank(),
        legend.position = "top")
```



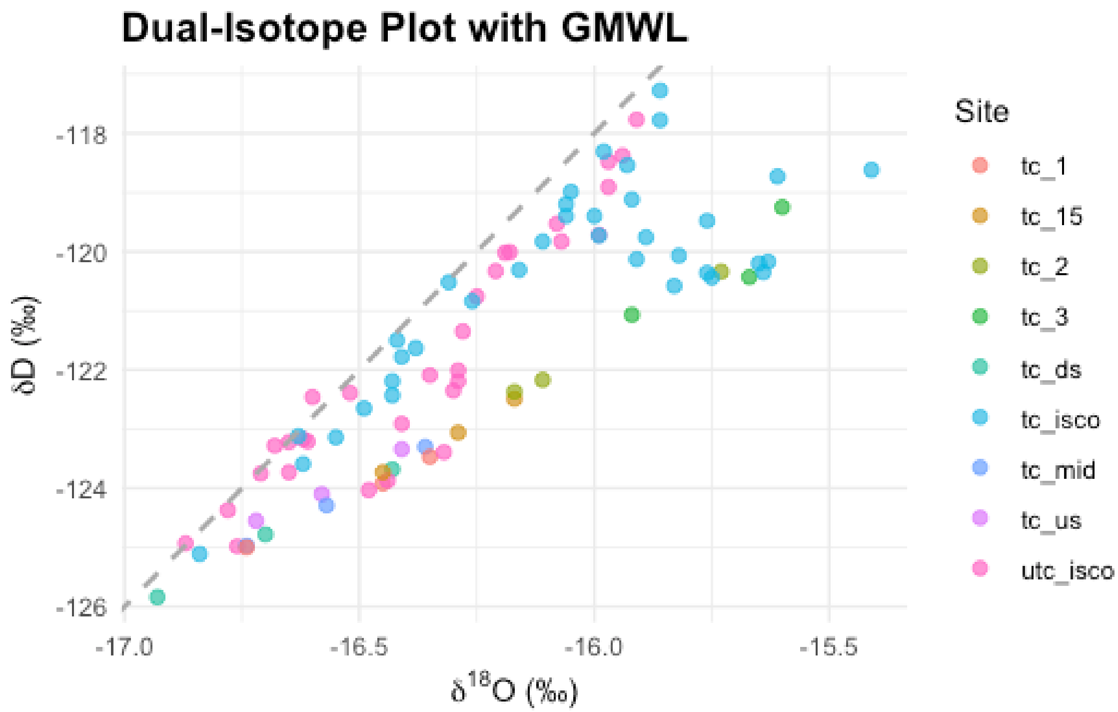
Time-Series of Line-Conditioned Excess (lc-excess)

```
ggplot(data = utc_isotope,
       aes(x = date,
           y = lc_excess)) +
  geom_hline(yintercept = 0,
             linetype = "dashed",
             color = "gray50",
             size = 0.8) +
  geom_point(size = 2.5,
             alpha = 0.8) +
  geom_line(size = 1,
            alpha = 0.7) +
  labs(title = "Time-Series of Line-Conditioned Excess (lc-excess)",
       subtitle = "Values below 0 indicate potential evaporative enrichment",
       x = "Sampling Date",
```

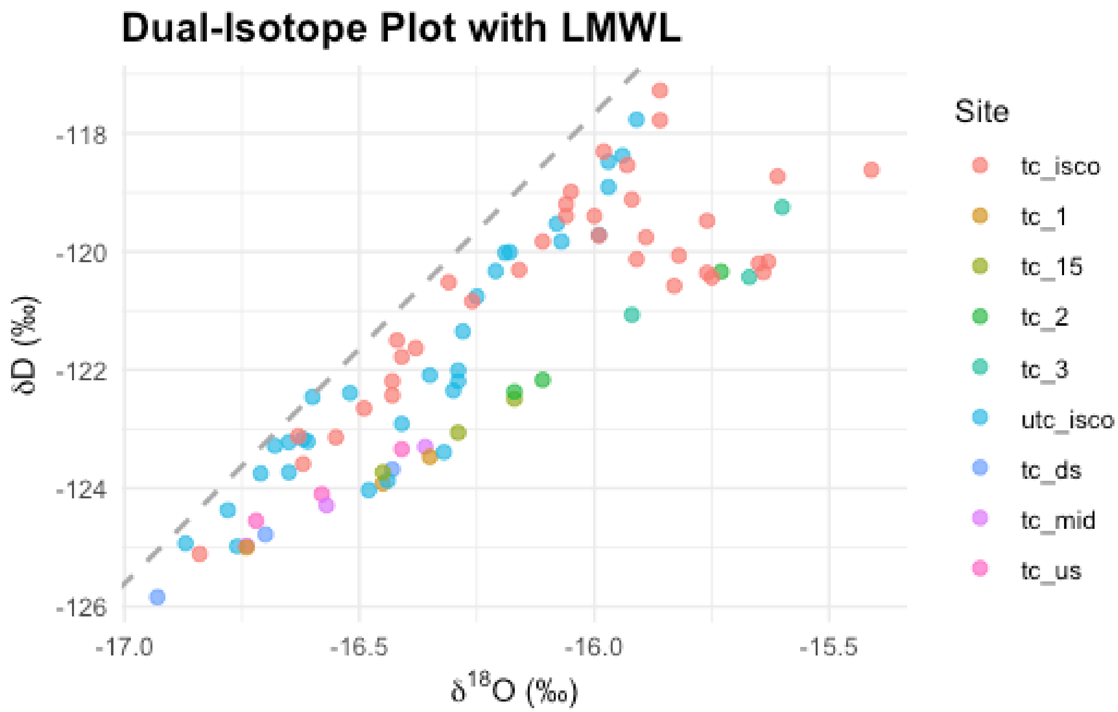
```
       y = "lc-excess (\u2030)",
       color = "Site") +
  ylim(-8,5) +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        panel.grid.minor = element_blank(),
        legend.position = "top")
```



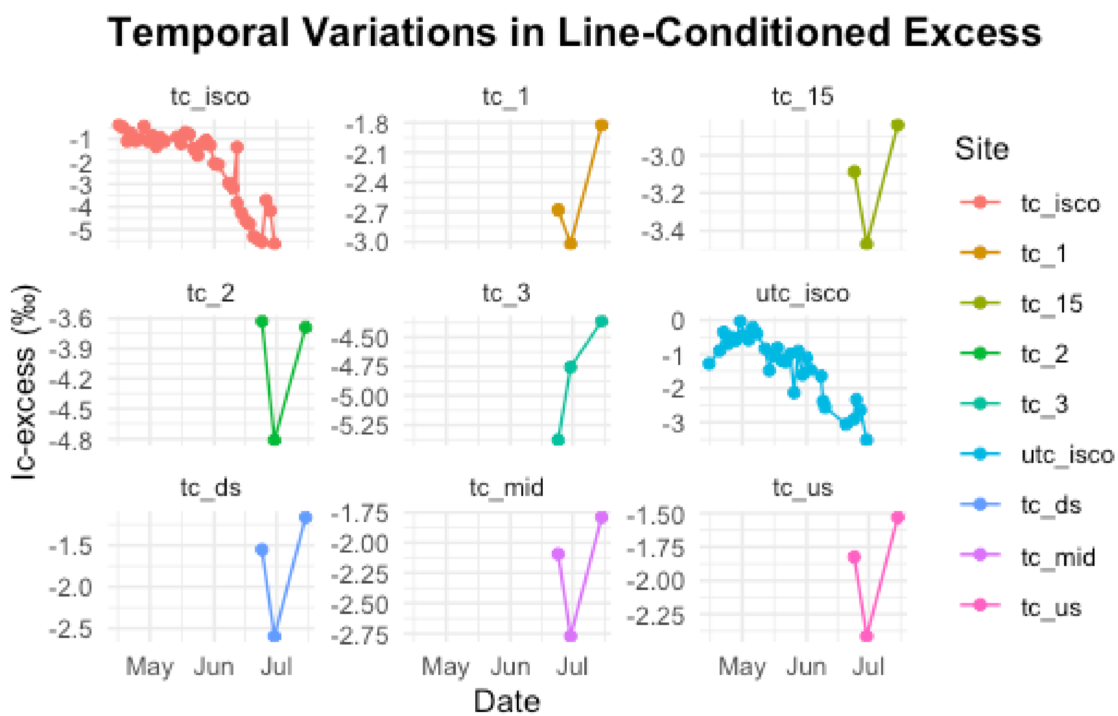

```
ggplot(compiled_isotope, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(alpha = 0.7, size = 2) +
  # Add Global Meteoric Water Line (GMWL: y = 8x + 10)
 geom_abline(intercept = 10, slope = 8, linetype = "dashed", color = "darkgray",
linewidth = 0.8) +
  labs(
    title = "Dual-Isotope Plot with GMWL",
    x = expression(paste(delta^{18}, "O (‰)")),
    y = expression(paste(delta, "D (‰)")),
    color = "Site"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))
```



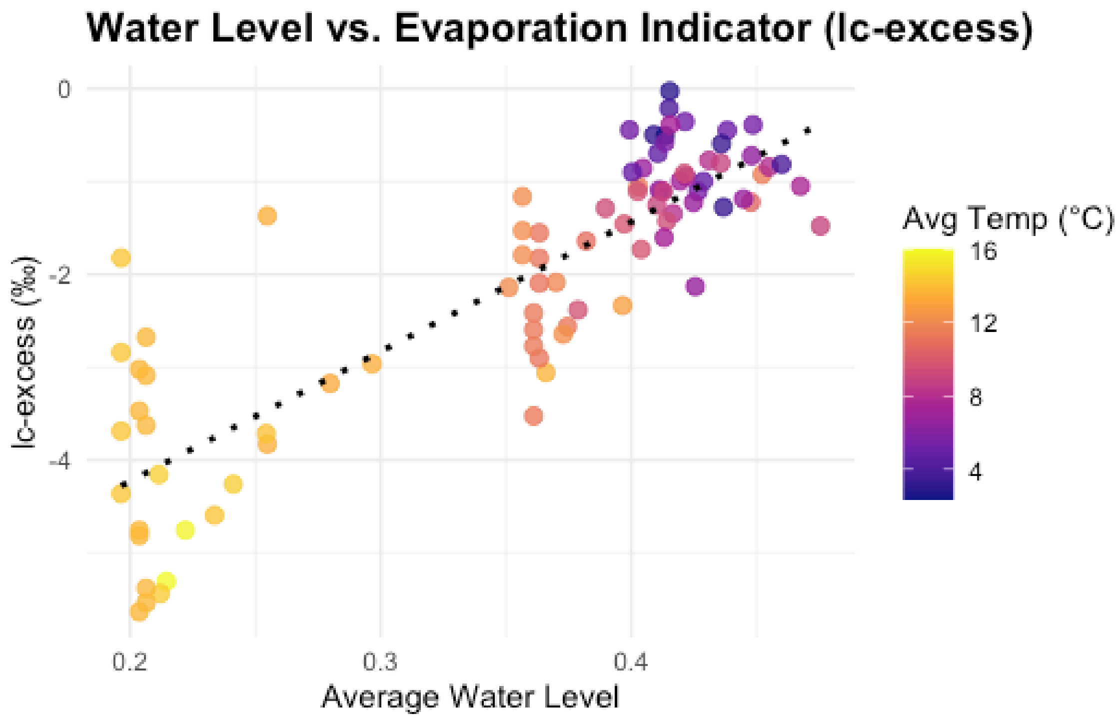

```r
ggplot(trail_creek, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(alpha = 0.7, size = 2) +
  # Add Local Mean Water Line (LMWL: y = 7.94 x + 9.37 )
   geom_abline(intercept  =  9.37,  slope  =  7.94,  linetype  =  "dashed",  color  =
"darkgray", linewidth = 0.8) +
  labs(
    title = "Dual-Isotope Plot with LMWL",
    x = expression(paste(delta^{18}, "O (‰)")),
    y = expression(paste(delta, "D (‰)")),
    color = "Site"
  ) +
  theme_minimal() +
   theme(plot.title = element_text(face = "bold", size = 14))
```




```
ggplot(trail_creek, aes(x = date, y = lc_excess, color = site)) +
  geom_line(linewidth = 0.5) +
  geom_point(size = 1.5) +
  labs(
    title = "Temporal Variations in Line-Conditioned Excess",
    x = "Date",
    y = "lc-excess (‰)",
    color = "Site"
  ) +
  scale_fill_viridis_d(option = "B") +
  facet_wrap(~site, scales = "free_y") +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))
```

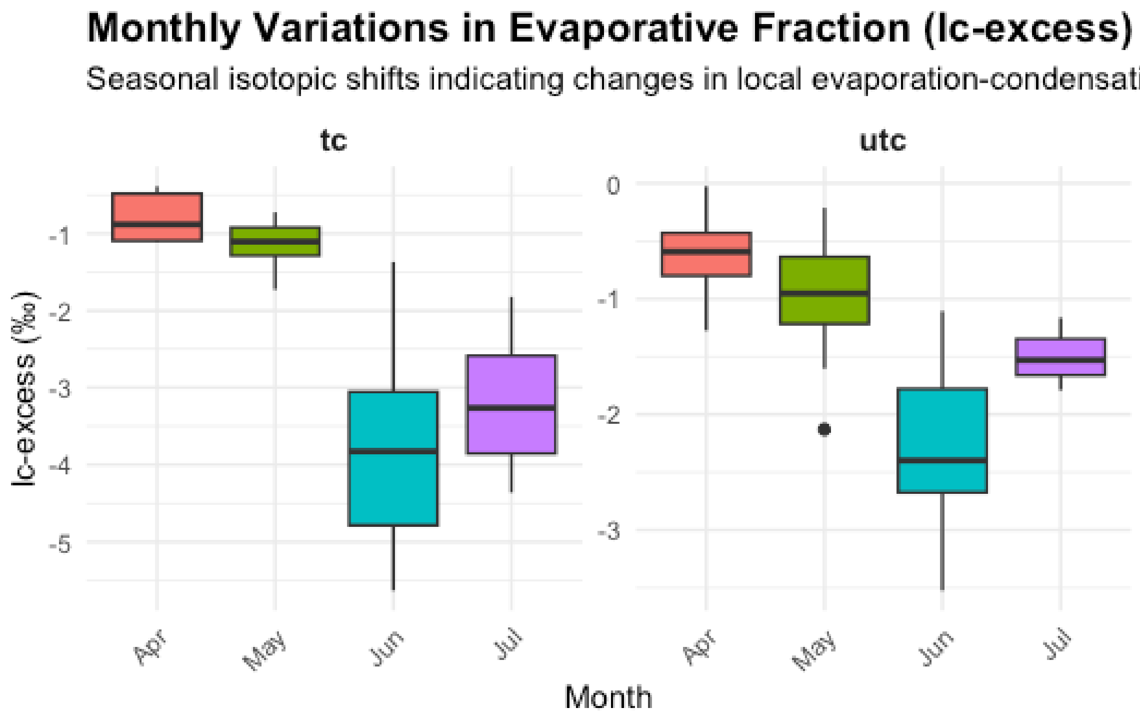



```
ggplot(isco_data, aes(x = avg_water_level, y = lc_excess)) +
  geom_point(aes(color = avg_temp), size = 2.5, alpha = 0.8) +
  geom_smooth(method = "lm", color = "black", linetype = "dotted", se = FALSE) +
  scale_color_viridis_c(option = "plasma") +  # Beautiful temperature gradient
  labs(
    title = "Water Level vs. Evaporation Indicator (lc-excess)",
    x = "Average Water Level",
    y = "lc-excess (‰)",
    color = "Avg Temp (°C)"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))
```



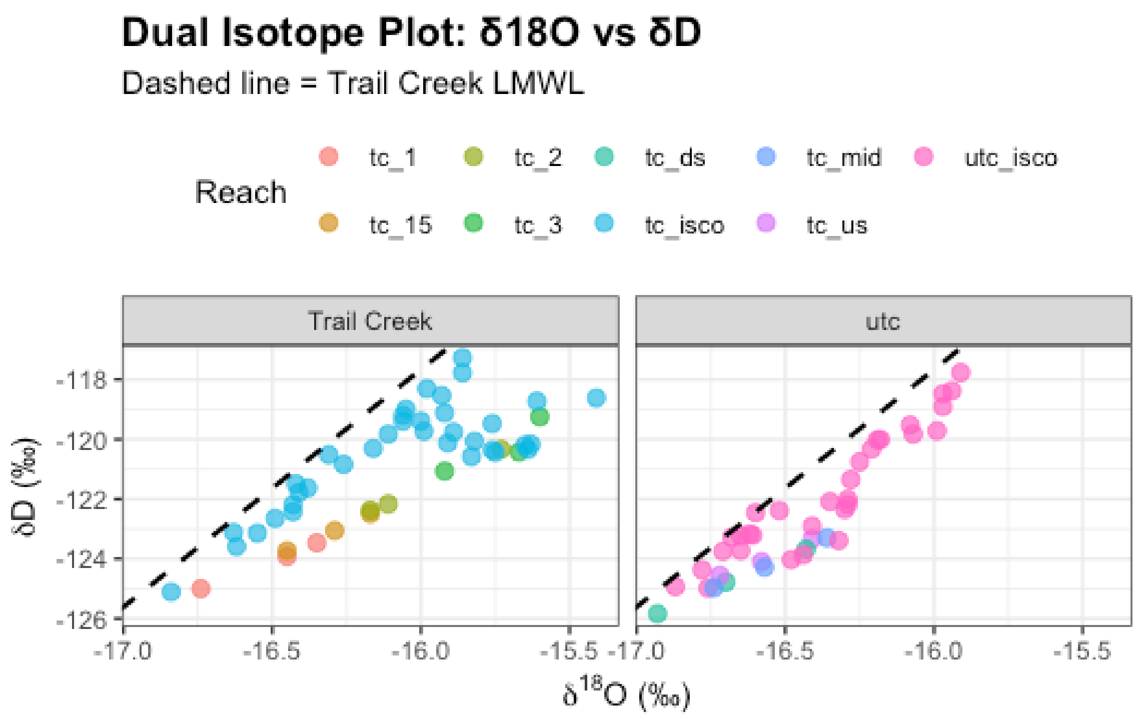
Monthly Variations in Evaporative Fraction (lc-excess) Across Study Sites

```
ggplot(trail_creek, aes(x = factor(month(date, label = TRUE, abbr = TRUE)), y =
lc_excess)) +
  geom_boxplot(aes(fill = factor(month(date))), show.legend = FALSE) +
  facet_wrap(~location, scales = "free_y") +
 labs(
      title = "Monthly Variations in Evaporative Fraction (lc-excess) Across
Study Sites",
    subtitle = "Seasonal isotopic shifts indicating changes in local evaporation-
condensation histories",
    x = "Month",
    y = "lc-excess (‰)"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14),
    strip.text = element_text(face = "bold", size = 11), # Emphasize site names
    axis.text.x = element_text(angle = 45, hjust = 1)   # Tilt month labels if
they overlap
  )
```



Dual-isotope plot (δ18O vs δD) with LMWL reference

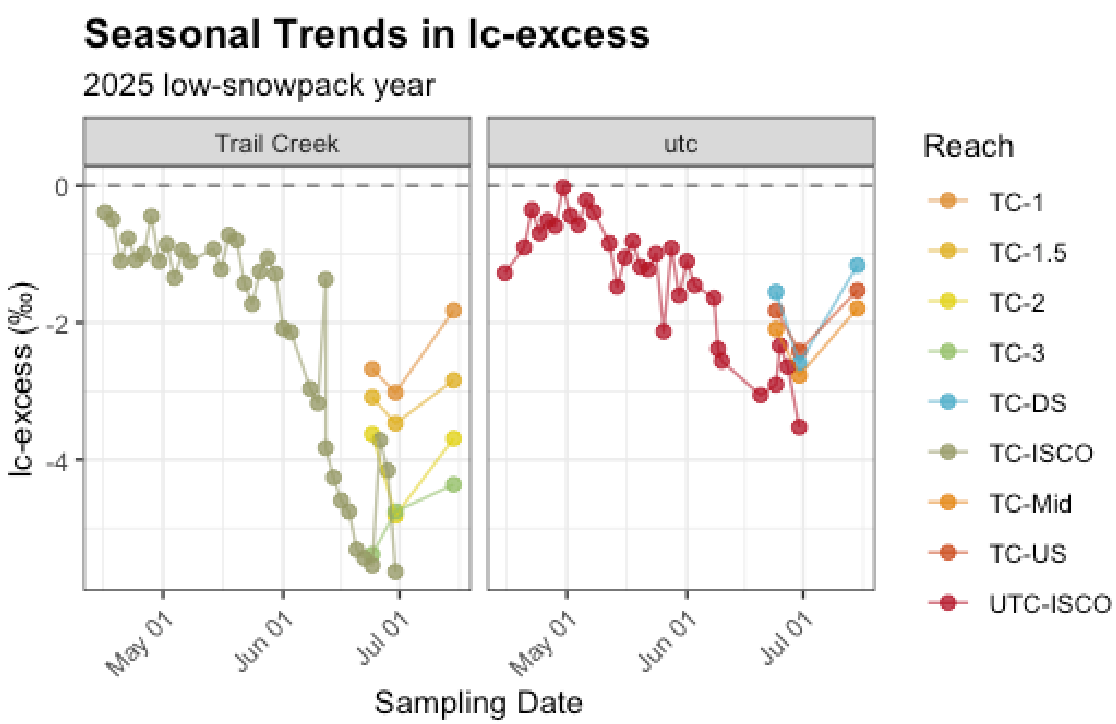
```
# Trail Creek LMWL calculated from OPIC

lmwl_slope <- 7.94
lmwl_intercept <- 9.37

ggplot(compiled_isotope, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(size = 2.5, alpha = 0.7) +
  geom_abline(slope = lmwl_slope, intercept = lmwl_intercept,
              linetype = "dashed", color = "black", linewidth = 0.8) +
  facet_wrap(~location, labeller = labeller(location = c(tc = "Trail Creek",
                                                          cc = "Coal Creek",
                                                             ic = "Italian
Creek"))) +
  labs(title = "Dual Isotope Plot: \u03b418O vs \u03b4D",
       subtitle = "Dashed line = Trail Creek LMWL",
       x = expression(paste(delta^18, "O (\u2030)")),
       y = expression(paste(delta, "D (\u2030)")),
       color = "Reach") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        legend.position = "top")
```



Time series of lc-excess across season, colored by reach, faceted by watershed

```
ggplot(compiled_isotope, aes(x = date, y = lc_excess, color = site, group =
interaction(site, location))) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
  geom_point(size = 2, alpha = 0.8) +
  geom_line(alpha = 0.6) +
  facet_wrap(~location, labeller = labeller(location = c(tc = "Trail Creek",
                                                          cc = "Coal Creek",
                                                       ic = "Italian Creek"))) +
  scale_x_date(date_labels = "%b %d") +
        scale_color_manual(values    =    wes_palette("FantasticFox1",    n    =
length(unique(compiled_isotope$site)),
                                         type = "continuous"),
                     labels = c("coal_1" = "CC-1", "coal_2" = "CC-2", "coal_3"
= "CC-3", "coal_4" =
                                 "CC-4", "coal_5" = "CC-5", "coal_6" = "CC-6",
"italian_1" = "IT-1",
                                   "italian_2" = "IT-2", "italian_3" = "IT-3",
"italian_4" = "IT-4",
                                   "italian_5" = "IT-5","tc_isco" = "TC-ISCO",
"tc_1" = "TC-1", "tc_15"
                                 = "TC-1.5", "tc_2" = "TC-2", "tc_3" = "TC-3",
```

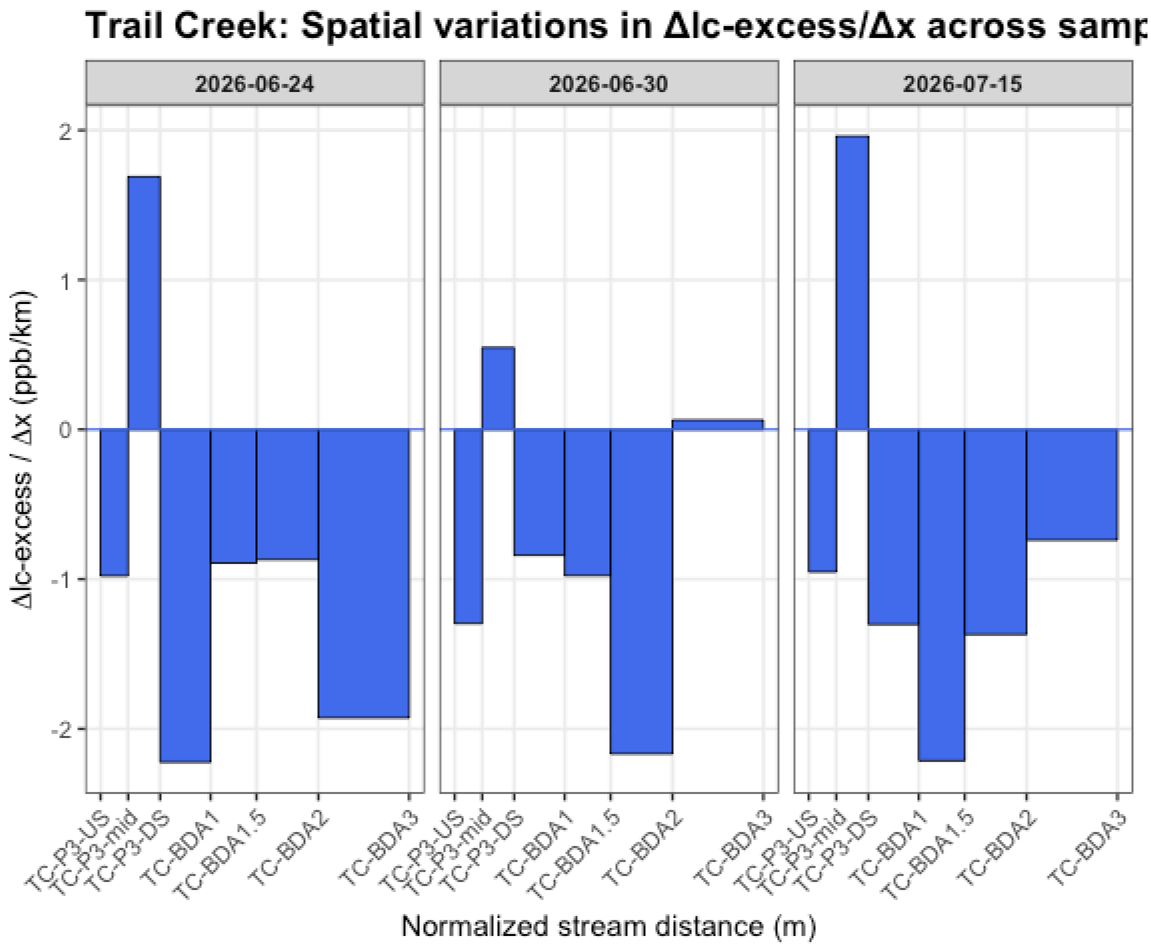
```
"tc_ds" = "TC-DS",
                              "tc_mid" = "TC-Mid", "tc_us" = "TC-US", "utc_isco"
= "UTC-ISCO")) +
  labs(title = "Seasonal Trends in lc-excess",
       subtitle = "2025 low-snowpack year",
       x = "Sampling Date", y = "lc-excess (\u2030)", color = "Reach") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        legend.position = "right",
        axis.text.x = element_text(angle = 45, hjust = 1))
```




```
# averaging the lc-excess and including standard error
delta_summary <- compiled_isotope |>
  mutate(site = factor(site, levels = site_order)) |>
  group_by(site) |>
  summarise(
    mean_lc = mean(lc_excess, na.rm = TRUE),
    se_lc = sd(lc_excess, na.rm = TRUE) / sqrt(n()),
    .groups = "drop"
  )

# creating a plot to show the mean lc-excess of each site
```

```
ggplot(delta_summary, aes(x = site, y = mean_lc, fill = site)) +
  geom_col(color = "black", width = 0.7) +
  geom_errorbar(aes(ymin = mean_lc - se_lc, ymax = mean_lc + se_lc),
                width = 0.3,
                color = "black") +
   scale_x_discrete(limits  =  c("tc_isco",  "tc_1",  "tc_15",  "tc_2",  "tc_3",
"utc_isco", "tc_ds", "tc_mid", "tc_us"),
  labels = c("tc_isco" = "TC-ISCO", "tc_1" = "TC-1", "tc_15" = "TC-1.5", "tc_2"
= "TC-2",
             "tc_3" = "TC-3", "tc_ds" = "IT-DS", "tc_mid" = "TC-Mid", "tc_us" =
"TC-US",
             "utc_isco" = "UTC-ISCO")) +
  scale_fill_viridis_d(option ="G") + # color palette
  theme_minimal() +
  labs(title = "Mean LC Excess by Site with Standard Error",
       x = "Sampling Site",
       y = "Mean LC Excess") +
  theme(plot.title = element_text(face = "bold", size = 14),
        axis.text.x = element_text(angle = 45, hjust = 1), # tilt labels
        legend.position = "none")
```

```r
# Buildings a look up table of stream distances
distances <- tibble(site = c("TC-P3-US", "TC-P3-mid", "TC-P3-DS", "TC-BDA1",
                             "TC-BDA1.5", "TC-BDA2", "TC-BDA3"),
                    dist_m = c(57.6, 333, 654, 1160, 1620, 2240, 3150)) |>
  mutate(xmax = dist_m,
        xmin = lag(dist_m, default = 0)) # previous site's distance, 0 for first

# Creating a filtered data frame for lc-excess
gradient_lc_clean <- gradient_tc |>
  select(site, normalized_stream_distance_m, `Δlc-excess/Δx_6-24`, `Δlc-excess/
Δx_6-30`, `Δlc-excess/Δx_7-15`) |>
   pivot_longer( cols  =  c(`Δlc-excess/Δx_6-24`,  `Δlc-excess/Δx_6-30`,  `Δlc-
excess/Δx_7-15`),
                names_to = "date",
                values_to = "gradient") |>
  mutate(date = case_when(date == "Δlc-excess/Δx_6-24" ~ "2026-06-24",
                          date == "Δlc-excess/Δx_6-30" ~ "2026-06-30",
                          date == "Δlc-excess/Δx_7-15" ~ "2026-07-15"),
         gradient = as.numeric(gradient)) |>
  left_join(distances %>% select(site, xmin, xmax), by = "site") |>
  mutate(site = factor(site, levels = distances$site)) |>
  filter(!is.na(gradient))

# Gradient plot
ggplot(gradient_lc_clean) +
  geom_rect(aes(xmin = xmin, xmax = xmax, ymin = 0, ymax = gradient),
            fill = "royalblue2", color = "black", linewidth = 0.3) +
  geom_hline(yintercept = 0, color = "royalblue2", linewidth = 0.4) +
  facet_wrap(~ date, nrow = 1) +
  scale_x_continuous(
    breaks = distances$dist_m,
    labels = distances$site) +
  labs(title = "Trail Creek: Spatial variations in Δlc-excess/Δx across sampling
dates",
       x = "Normalized stream distance (m)",
       y = expression(paste(Delta, "lc-excess / ", Delta, "x (ppb/km)"))) +
  theme_bw(base_size = 11) +
  theme(plot.title = element_text(face = "bold", size = 14),
        panel.grid.minor = element_blank(),
        strip.text = element_text(face = "bold"),
        axis.text.x = element_text(angle = 45, hjust = 1))
```



All reaches show lc-excess decreasing downstream, on both dates. Since lc-excess becoming more negative = more evaporative enrichment, this version says every single reach along Trail Creek is adding evaporative signal, with no recovery zones. That's a materially different story than "some reaches enrich, some reaches dilute" — it now reads as continuous, cumulative evaporative loss along the whole restored gradient.

Where the deepest enrichment is happening differs by date:

6-24: TC-P3-DS reach is the sharpest dip (~−2.2), essentially a local hotspot just downstream of the mid-P3 restoration point. 6-30: TC-BDA1.5→BDA2 is now the sharpest dip (~−2.15), while TC-P3-DS has relaxed to a much smaller magnitude (~−0.55) compared to 6-24.

TC-BDA3 is the outlier reach on both dates — it's close to flat (near-zero gradient), meaning that final reach adds almost no additional evaporative signal, regardless of date. That's consistent between both panels and is probably your most robust finding here: whatever's happening upstream, the lowest reach isn't contributing much additional fractionation.

```r
gradient_cum_tc <- gradient_tc |>
  select(site, cumulative_dams,
         `Δlc-excess/Δcum_dam_length_6-24`, `Δlc-excess/Δcum_dam_length_6-30`,
`Δlc-excess/Δcum_dam_length_7-15`) |>
  pivot_longer(
              cols    =    c(`Δlc-excess/Δcum_dam_length_6-24`,    `Δlc-excess/
Δcum_dam_length_6-30`, `Δlc-excess/Δcum_dam_length_7-15`),
    names_to = "date",
    values_to = "gradient"
  ) |>
  mutate(
    date = case_when(
      date == "Δlc-excess/Δcum_dam_length_6-24" ~ "2026-06-24",
      date == "Δlc-excess/Δcum_dam_length_6-30" ~ "2026-06-30",
      date == "Δlc-excess/Δcum_dam_length_7-15" ~ "2026-07-15"
    ),
    gradient = as.numeric(gradient),
    cumulative_dams = as.numeric(cumulative_dams),
    site = factor(site, levels = c("TC-P3-US", "TC-P3-mid", "TC-P3-DS",
                                  "TC-BDA1", "TC-BDA1.5", "TC-BDA2", "TC-BDA3"))
  ) |>
  filter(!is.na(gradient))


ggplot(gradient_cum_tc, aes(x = site, y = gradient, fill = site)) +
  geom_col(color = "black", width = 0.7) +
  geom_hline(yintercept = 0, color = "steelblue", linewidth = 0.4) +
  facet_wrap(~ date, nrow = 1) +
  scale_fill_viridis_d(option = "G") +
  scale_x_discrete(labels = function(x) {
    lookup <- setNames(gradient_cum_tc$cumulative_dams, gradient_cum_tc$site)
    lookup[x]
  }) +
  labs(
  x = "Cumulative dam count",
   y  =  expression(Delta  *  "lc-excess/"  *  Delta  *  "cumulative  dam  length  ("  *
"\u2030" * ")/m")
) +
  theme_bw(base_size = 12) +
  theme(
    strip.text = element_text(face = "bold", size = 11),
    strip.background = element_rect(fill = "gray85", color = NA),
    panel.grid.minor = element_blank(),
```

```
    panel.grid.major.x = element_blank(),
    legend.title = element_text(face = "bold")
  )
```